R-INVARIANT AFFINE THICKENING PAIRS AROUND GENERIC ORBIT POINTS FOR LOCALLY NOETHERIAN U WITH FINITE SOURCE AND TARGET

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Allowed standard prerequisites (editorial).

- The R-invariant thickening pair W' inside W with finite locally free restriction and the covering property (more-groupoids chapter, Lemma lemma-finite-flat-over-almost-dense-subscheme, canonical tag 0ABA; collected on this lane as L22511). cited at proof lines 2861-2862 and used through lines 2863-2874. Provides the pair W' inside W whose properties drive the orbit-containment and the replacement reduction; its covering (2) puts some u_{ji} in the closure of W.
- Images of affine schemes by finite morphisms are affine, Noetherian setting (cohomology of schemes chapter, Lemma coherent-lemma-image-affine-finite-morphism-affine-Noetherian, canonical tag 01YQ). cited at proof lines 2868-2872. Reduces the task to finding an R-invariant affine open of W': the corresponding V inside W is then affine since W' to W is finite.
- Affine opens containing finitely many maximal-type points of a quasi-separated scheme (properties chapter, Lemma properties-lemma-maximal-points-affine, canonical tag 01ZV). cited at proof lines 2877-2878. Finds an affine open containing the finitely many orbit points in the restricted, quasi-separated situation.
- A locally Noetherian scheme is quasi-separated (properties chapter, Lemma properties-lemma-locally-Noetherian-quasi-separated, canonical tag 01OY). cited at proof lines 2879-2880. Justifies the applicability of the maximal-points lemma to the locally Noetherian U (and its opens).
- Existence of R-invariant affine opens containing finitely many points for groupoids with finite locally free structure maps (groupoids chapter, Lemma groupoids-lemma-find-invariant-affine, canonical tag 03JE). cited at proof lines 2881-2883. Produces the final R-invariant affine open containing the orbit points in the finite locally free restriction.

Source reference. Original lemma 0ABA: sources/stacks-chow/tags.txt, source lines 13013–13013. Cited at proof lines 2861-2862: supplies the R-invariant pair W'

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inside W, its covering property (2), and the finite locally free restriction used in the reduction.

Source reference. Original lemma 01YQ: sources/stacks-chow/tags.txt, source lines 2565–2565. Cited at proof lines 2868-2872: reduces the construction to an R-invariant affine open of W', the thickening V being affine by finiteness.

Source reference. Original lemma 01ZV: sources/stacks-chow/tags.txt, source lines 2610–2610. Cited at proof lines 2877-2878: finds an affine open containing the finitely many orbit points.

Source reference. Original lemma 01OY: sources/stacks-chow/tags.txt, source lines 2211–2211. Cited at proof lines 2879-2880: a locally Noetherian scheme is quasi-separated, the hypothesis needed for the maximal-points lemma.

Source reference. Original lemma 03JE: sources/stacks-chow/tags.txt, source lines 4636–4636. Cited at proof lines 2881-2883: produces the R-invariant affine open containing the orbit points for a groupoid with finite locally free structure maps.

Lemma 0.2. *Let (U, R, s, t, c) be a groupoid scheme over a scheme S. Assume s, t finite, U is locally Noetherian, and $u_1, \ldots, u_m \in U$ points whose orbits consist of generic points of irreducible components of U. Then there exist R-invariant subschemes $V' \subset V \subset U$ such that*

1. *$u_1, \ldots, u_m \in V'$,*
2. *V is open in U,*
3. *V' and V are affine,*
4. *$V' \subset V$ is a thickening,*
5. *the morphisms s', t' of the restriction (V', R', s', t', c') are finite locally free.*

Proof. Let $\{u_{j1}, \ldots, u_{jn_j}\}$ be the orbit of u_j. Let $W' \subset W \subset U$ be as in Lemma 0ABA. Since $U = t(s^{-1}(\overline{W}))$ we see that at least one $u_{ji} \in \overline{W}$. Since u_{ji} is a generic point of an irreducible component and U locally Noetherian, this implies that $u_{ji} \in W$. Since W is R-invariant, we conclude that $u_j \in W$ and in fact the whole orbit is contained in W. By Cohomology of Schemes, Lemma 01YQ it suffices to find an R-invariant affine open subscheme V' of W' containing $u_1, \ldots, u_m$ (because then we can let $V \subset W$ be the corresponding open subscheme which will be affine). Thus we may replace (U, R, s, t, c) by the restriction (W', R', s', t', c') to W'. In other words, we may assume we have a groupoid scheme (U, R, s, t, c) whose morphisms s and t are finite locally free. By Properties, Lemma 01ZV we can find an affine open containing $\{u_{ij}\}$ (a locally Noetherian scheme is quasi-separated by Properties, Lemma 01OY). Finally, we can apply Groupoids, Lemma 03JE to conclude. □